

Booking Automation

Description

This project makes setting up meetings easy by letting users pick, change, or cancel their schedule online. It automatically updates the calendar and sends an email with a PDF summary of all the details for every booking to be organized and professional, using n8n to automate the entire process.

Setup

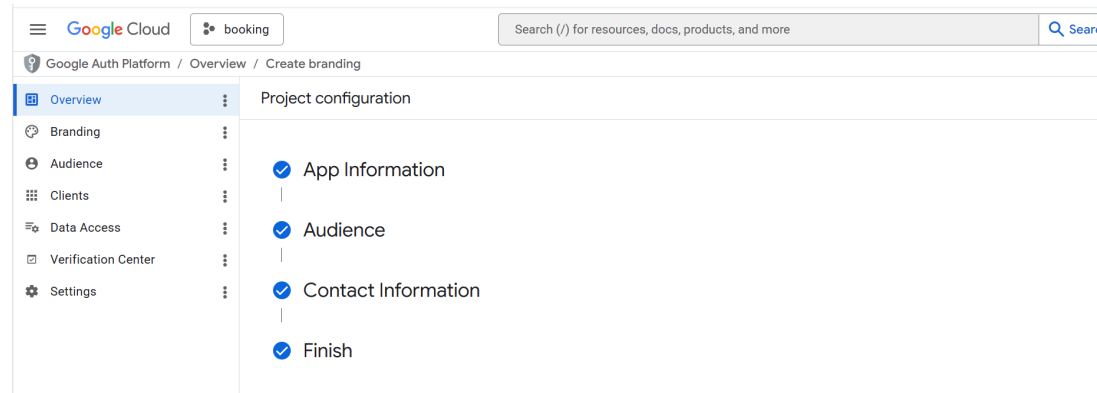
Prerequisites

n8n	Self-hosted or Sign-up to https://n8n.partnerlinks.io/q3whjtr3e9f9
Docker Desktop	1. Create folder where you're gonna store the data and workflows under n8n. Open terminal (or PowerShell/Command Prompt) and run: • <code>mkdir n8n-docker-setup && cd n8n-docker-setup</code> 2. Install Docker • <code>sudo apt update && sudo apt install docker.io docker-compose-v2 -y</code> 3. Create the data directories and configuration files. Run these commands inside project folder: • <code>mkdir n8n_data postgres_data</code> 4. Automatically create the .env and yaml. Note that you will need to configure the necessary details here including APIs and credentials. • <code>touch .env docker-compose.yaml</code> 5. Open the main folder and inside it is the <code>docker-compose.yaml</code>. Add the code below on

	that file. Configure the <code>.env</code> as well. <pre>Run Service gotenberg: image: gotenberg/gotenberg:8 restart: always ports: - "3010:3000" networks: - n8n_net</pre>
Gotenberg	For self-hosted, use Gotenberg as a free and unlimited PDF generator. - Open Docker Desktop app - Open command prompt. Set up Gotenberg by running it on command prompt and check on Docker Desktop for confirmation. <code>docker run --rm -p 3010:3000 gotenberg/gotenberg:8</code> - When everything is good, start by going to terminal and run <code>docker start gotenberg</code> <code>docker ps</code> - When the port for Gotenberg is already showing, it's working. - Paste the ngrok link on browser to start creating workflows.

Google Cloud API

Create Client ID and Client Token by filling up this Project Configuration in Google Cloud Console.



Now this is where the Client ID is being defined for the workflow to work.

Google Cloud

booking

Search (/) for resources, docs, products, &

Google Auth Platform / Clients / Create client

Overview

Branding

Audience

Clients

Data Access

Verification Center

Settings

Create OAuth client ID

A client ID is used to identify a single app to Google's OAuth servers. If your app runs on multiple platforms, each will need its own client ID. See [Setting up OAuth 2.0](#) for more information. [Learn more](#) about OAuth client types.

Application type *
Web application

Name *
Booking

The name of your OAuth 2.0 client. This name is only used to identify the client in the console and will not be shown to end users.

The domains of the URIs you add below will be automatically added to your [OAuth consent screen](#) as [authorized domains](#).

Authorized JavaScript origins

For use with requests from a browser

+ Add URI

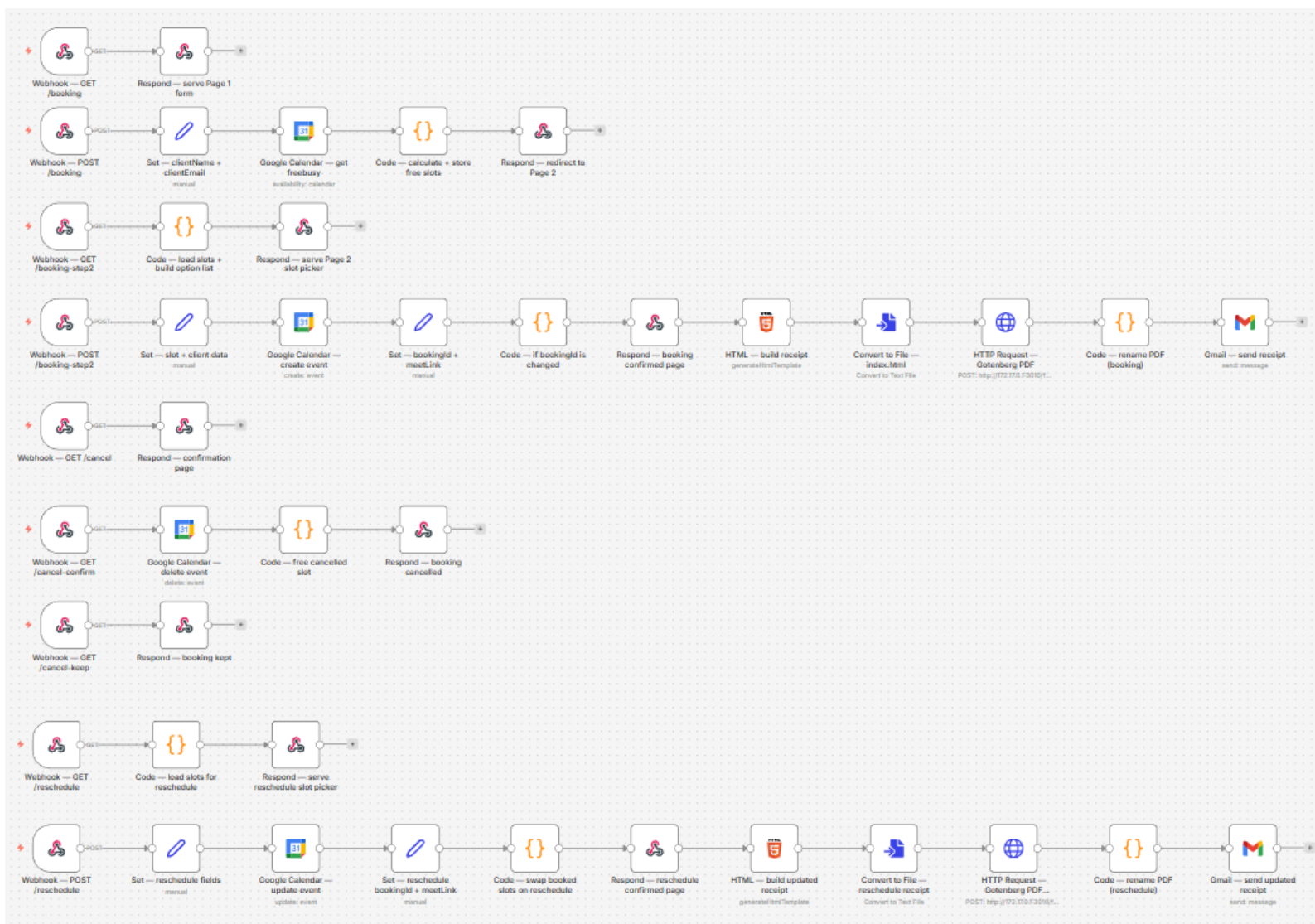
Authorized redirect URIs

For use with requests from a web server

URIs 1 *
<https://bradford-antral-mortally.ngrok-free.dev/rest/oauth2-credential/>

+ Add URI

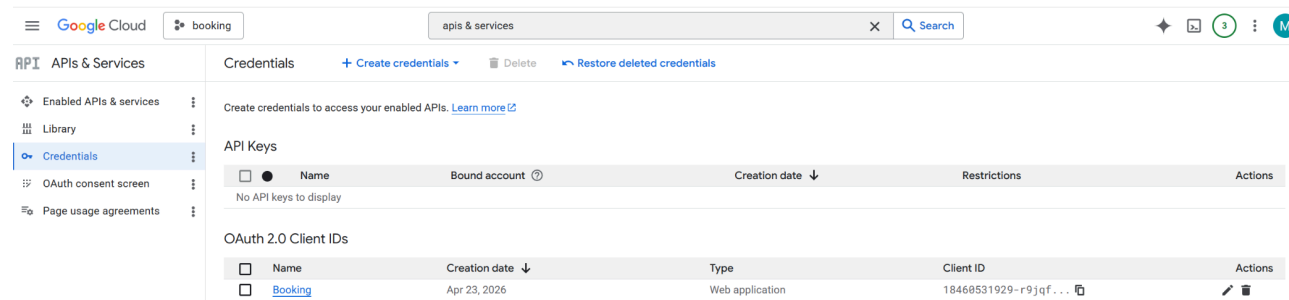
n8n Workflow



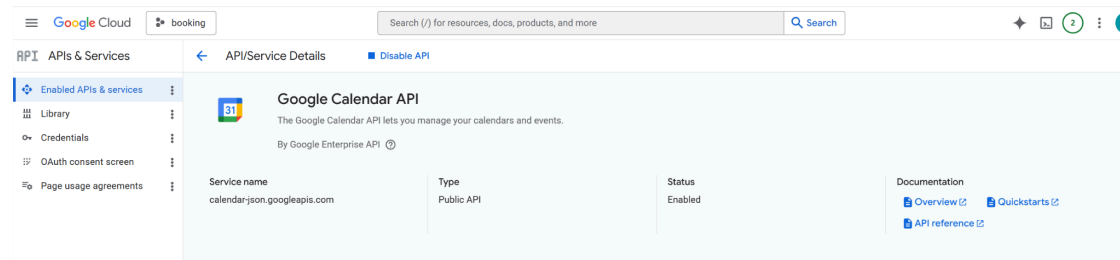
n8n Credentials Setup

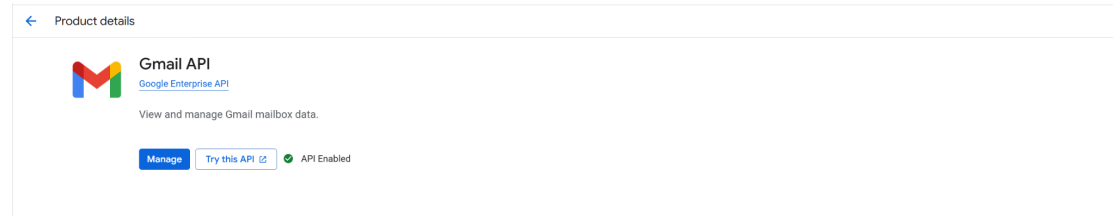
Gmail Oauth2 API

1. Open the Google Cloud Console through <https://console.cloud.google.com/>
2. Set up Credentials on the Menu
 - a. Go to APIs and Services > Credentials > OAuth client ID, and get both the Client ID and Client secret to be pasted in **Sheets** node.
 - b. Add the OAuth Redirect URL in Cloud Console's **Authorised redirect URLs** indicated in the n8n Sheets node.

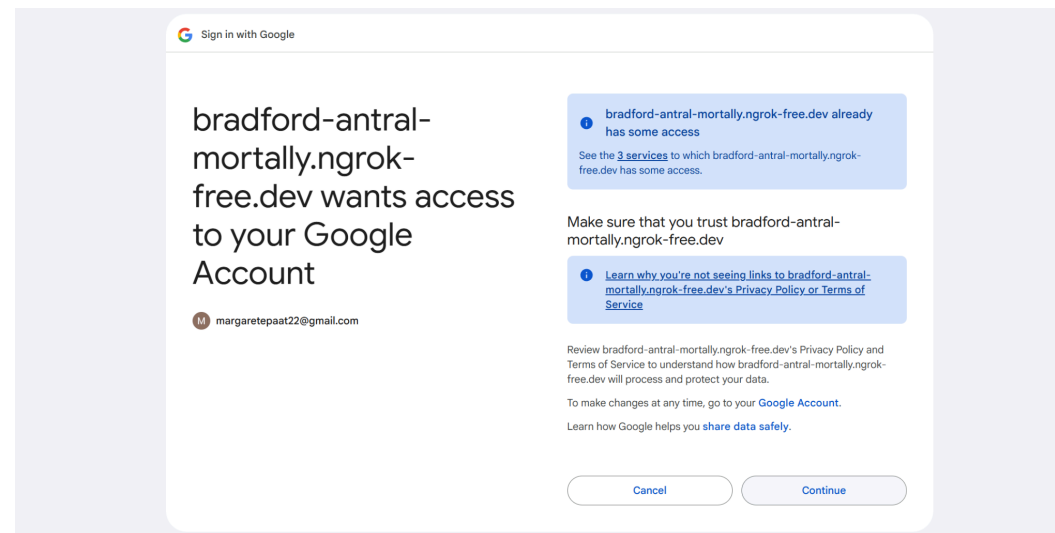


3. Set up API
 - a. Go to APIs and Services > Enabled API and Services.
 - b. Make sure **Gmail, Google Calendar, Google Cloud APIs** are enabled





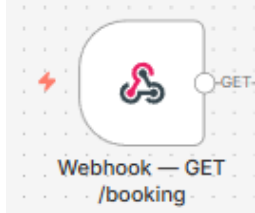
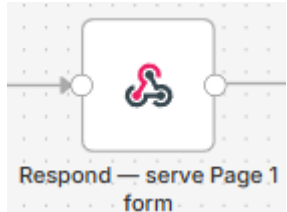
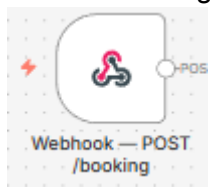
4. **Save the details and sign in with a Gmail account. Tick the checkboxes based on needed information.**

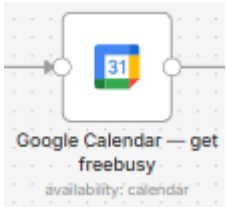


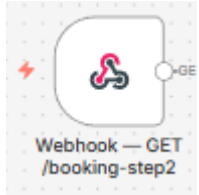
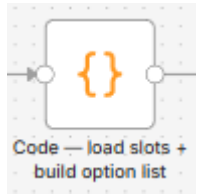
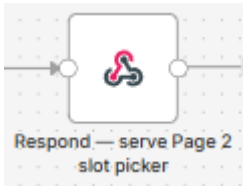
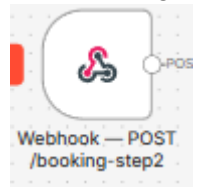
5. **Do the same with the Google Calendar.**

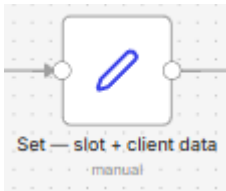
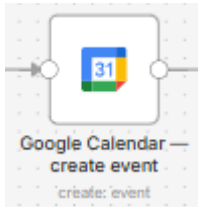
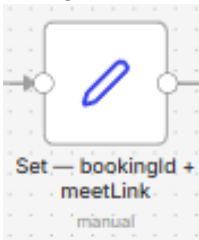
n8n Workflow Setup

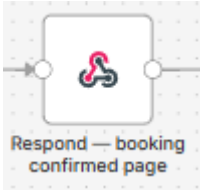
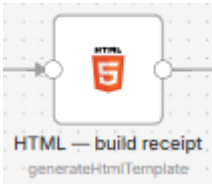
- Create a New Workflow

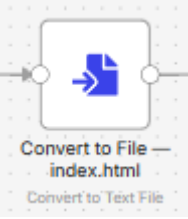
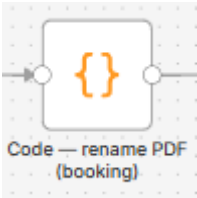
Node	Configuration
1. GET booking 	→ Uses GET operation so that when the form is being opened, it can read the data • Production URL sample: https://ngrok-free.dev/webhook/booking • HTTP Method : GET • Path : booking • Authentication : None • Respond : USing 'Respond to Webhook' Node
1.2 Respond Page 1 form 	→ The HTML that responds, basically first form design or placeholder of the contents • Respond With : Text • Response Body : What you want for the form to look like. It fetches client's name and email of the booker • Response Headers - Name : Content-Type - Value : text/html; charset=utf-8
2. POST booking 	→ Prepares to send the data to the next form • Production URL sample: https://ngrok-free.dev/webhook/booking • HTTP Method : POST • Path : booking • Authentication : None • Respond : Using 'Respond to Webhook' Node
2.2 Set name and email 	→ Sets the client name and email to json • Mode : Manual Mapping • Fields to Set - name : clientName, value : {{ \$json.body.name }} (String) - name : clientEmail, value : {{ \$json.body.email }} (String) • Include Other Input Fields : On • Input Fields to Include : All

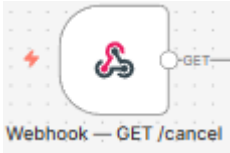
2.3 Google Calendar - get days 	→ Checking for dates for the next 7 days • Credential : Google Calendar account • Resource : Calendar • Operation : Availability • Calendar : From list; primary • Start Time : {{ \$now.toISO() }} • End Time : {{ \$now.plus({days:7}).toISO() }}
2.4 Code - calculate 	→ This script identifies available 30-minute meeting slots over the next week by filtering out weekends, Philippine holidays, lunch breaks, and existing calendar conflicts. It generates a list of free timestamps between 8:00 AM and 5:00 PM, then saves these options into n8n's memory indexed by the client's email for easy retrieval. • Mode : Run Once for All Items • Language : Javascript • Javascript : computes the details above
2.5 Responds, goes to Page 2 	→ Webhook that responds after a button is clicked • Respond With : Redirect • Redirect URL (Expression) : <pre> {{ 'https://' + \$('Set - clientName + clientEmail').item.json.headers['host'] + '/webhook/booking-step2?name=' + encodeURIComponent(\$json.clientName) + '&email=' + encodeURIComponent(\$json.clientEmail) }} </pre> • Response Code : 302

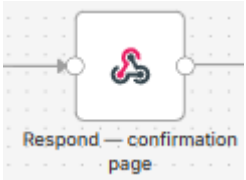
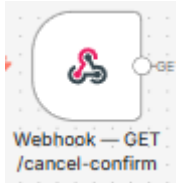
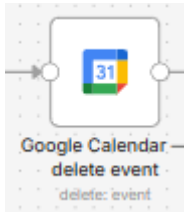
3. Get booking step 2 	→ Uses GET operation so that when the form is being opened, it can read the data for step 2 • Production URL sample: https://ngrok-free.dev/webhook/booking-2 • HTTP Method : GET • Path : booking-step2 • Authentication : None • Respond : Using 'Respond to Webhook' Node
3.2 Code for losing slots 	→ This script pulls the saved slots from memory to show them to the client in a dropdown menu. It checks the client's email, removes any slots that were already booked by others, and converts the technical timestamps into a readable format for a web form. • Mode : Run Once for All Items • Language : Javascript • Javascript : computes the details above
3.3 Webhook Respond slot picker 	→ This is the form or design that client sees after seeing the available slots in a dropdown. This step is for booking confirmation • Respond With : Text • Response Body : javascript for UI interface • Response Headers - Name : Content-Type - Value : text/html; charset=utf-8
4. POST booking-step2 	→ This prepares the data retrieved after confirmation • Production URL sample: https://ngrok-free.dev/webhook/booking-step2 • HTTP Method : POST • Path : booking-step2 • Authentication : None • Respond : Using 'Respond to Webhook' Node
4.2 Set	→ Continue to retain the client name and email, plus the selected time slot

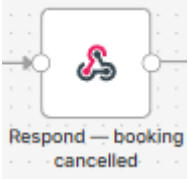
 Set — slot + client data manual	• Mode : Manual Mapping • Fields to Set - name : clientName, value : {{ \$json.body.clientName }} (String) - name : clientEmail, value : {{ \$json.body.clientEmail }} (String) - name : selectedSlot, value : {{ \$json.body.selectedSlot }} (String)
4.3 Google Calendar - create event  Google Calendar — create event create: event	→ Continue to retain the client name and email, plus the selected time slot • Credential : Google Calendar account • Resource : Event • Operation : Create • Calendar : From list, primary • Start : {{ \$json.selectedSlot }} • End : {{ DateTime.fromISO(\$json.selectedSlot).plus({minutes:30}).toISO() }} • Use Default Reminds : On • Additional Fields - Attendees : {{ \$json.clientEmail }} - Conference Data : Google Meet - Summary : Booking — {{ \$json.clientName }}
4.4 Setting additional info  Set — bookingId + meetLink manual	→ Set booking ID, meet link, to send it later • Mode : Manual Mapping • Fields to Set - name : bookingID, value : {{ \$json.id }} (String) - name : meetLink, value : {{ \$json.hangoutLink }} (String) - name : clientName, value : {{ \$('Set — slot + client data').item.json.clientName }} (String) - name : clientEmail, value : {{ \$('Set — slot + client data').item.json.clientEmail }} (String) - name : selectedSlot, value : {{ \$('Set — slot + client data').item.json.selectedSlot }} (String)
4.5 Code - if booking ID is changed	→ Handles when booking ID is changed

	• Mode : Run Once for All Items • Javascript <pre>const staticData = \$getWorkflowStaticData('global'); const email = \$json.clientEmail; const slot = \$json.selectedSlot; if (staticData[email]) { staticData[email].bookingId = \$json.bookingId; staticData[email].selectedSlot = slot; } if (!staticData._bookedSlots) staticData._bookedSlots = []; if (!staticData._bookedSlots.includes(slot)) { staticData._bookedSlots.push(slot); } return [{ json: \$json }];</pre>
4.6 Respond - booking confirmed 	→ This finally says that the booking is successfully confirmed and automatically added to calendar. • Respond With : Text • Response Body: • Response Headers <pre>Name : Content-Type Value : text/html; charset=utf-8</pre>
4.7 HTML - build receipt 	→ This is the design of the pdf that will be used as a receipt. Aside from booking details, the header also contains agency name, location, and email. • Operation : Generate HTML Template
4.8 Convert to File	→ This defines the content layout and appearance to be passed.

 Convert to File — index.html Convert to Text File	• Operation : Convert to Text File • Text Input Field : html • Put Output File in Field : data • Options File Name : index.html
4.9 HTTP Request Gotenberg  HTTP Request — Gotenberg PDF POST: http://172.17.0.1:3010/f...	→ This connects to the Gotenberg for pdf generation. • Method : POST • URL : http://<ip address>/forms/chromium/convert/html • Authentication : None • Send Body : On • Body Content Type : Form-Data • Body Type : n8n Binary File Name : files Input Data Field Name : data • Response Format : File • Put Output Data : data
4.10 Code - rename PDF  Code — rename PDF (booking)	→ This creates a personalized pdf name for the client so it won't generate any pdf name. • Mode : Run Once for All Items • Language : JavaScript • Javascript: <pre>const nameRaw = \$('Set - bookingId + meetLink').item.json.clientName 'Client'; const parts = nameRaw.trim().split(' '); const firstName = parts[0] 'Firstname'; const lastName = parts.slice(1).join(' ') 'Lastname'; const fileName = `\${lastName}_\${firstName}_Booking.pdf`; const binaryKey = Object.keys(\$binary)[0]; const binaryData = \$binary[binaryKey]; binaryData.fileName = fileName;</pre>

	<pre>binaryData.mimeType = 'application/pdf'; return [{ json: \$json, binary: { data: binaryData } }];</pre>
4.11 Gmail - send receipt 	→ This creates both message and pdf receipt to the email of the client. • Credential: Gmail account 2 • Resource : Message • Operation : Send • To:{{ \$('Set - bookingId + meetLink').item.json.clientEmail }} • Subject : Your booking is confirmed • Email Type : HTML • Message : <pre> {{ \$('Set - bookingId + meetLink').item.json.clientName }}, Your booking is confirmed. Your receipt PDF is attached and you can join Google Meet: {{ \$('Set - bookingId + meetLink').item.json.meetLink }}. If you want to cancel or reschedule, you may do so by opening the attached PDF. See you soon!</pre> • Options : <pre> Attachments : data</pre>
5. Webhook GET/cancel 	→ Uses GET operation so that when the form is being opened, it can read the data • Production URL sample: https://ngrok-free.dev/webhook/cancel • HTTP Method : GET • Path : cancel • Authentication : None • Respond : USing 'Respond to Webhook' Node
5.2 Respond - confirmation page	→ This is the HTML form that will show after pressing the cancel button in pdf.

	• Respond With : Text • Response Body : What you want for the form to look like. It fetches client's name and email of the booker • Response Headers - Name : Content-Type - Value : text/html; charset=utf-8
6. Webhook - GET/cancel-confirm 	→ Reads data if client wants to keep the booking. • Production URL sample: https://ngrok-free.dev/webhook/cancel-confirm • HTTP Method : GET • Path : cancel-confirm • Authentication : None • Respond : USING 'Respond to Webhook' Node
6.2 Google Calendar - delete event 	→ Checks for the event ID for the delete option. • Credential : Google Calendar account • Resource : Event • Operation : Delete • Calendar : From list; primary • Event ID : {{ \$json.query.id }}
6.3 Code - free cancelled slot 	→ Identifies the cancelled slots and letting the availability schedule be seen • Mode : Run Once for All Items • Language : Javascript • Javascript : computes the details above
6.4 Respond - booking cancelled	→ The design after the booking is successfully cancelled to show that the booking schedule is deleted.

	• Respond With : Text • Response Body : What you want for the form to look like • Response Headers - Name : Content-Type - Value : text/html; charset=utf-8
7. Webhook - GET/cancel-keep	→ Uses GET operation so that when the form is being opened, it can keep the data • Production URL sample: https://ngrok-free.dev/webhook/cancel-keep • HTTP Method : GET • Path : cancel-keep • Authentication : None • Respond : USING 'Respond to Webhook' Node
7.2 Respond - booking kept	→ Assures the client that the booking is not cancelled. The booking is then kept. • Respond With : Text • Response Body : What you want for the form to look like. • Response Headers - Name : Content-Type - Value : text/html; charset=utf-8
8. Webhook - GET/reschedule	→ Uses GET operation for the rescheduling. • Production URL sample: https://ngrok-free.dev/webhook/reschedule • HTTP Method : GET • Path : reschedule • Authentication : None • Respond : USING 'Respond to Webhook' Node
8.2 Code - load slots for reschedule	→ This shows the available slots after rescheduling. • Mode : Run Once for All Items • Language : Javascript • Javascript : computes the details above

8.3 Respond - serve reschedule slot picker	→ Assures the client that the booking is not cancelled. The booking is then kept. - Respond With : Text - Response Body : What you want for the form to look like. - Response Headers - Name : Content-Type - Value : text/html; charset=utf-8
9. Webhook - POST/reschedule	→ Prepares to send the data for rescheduling - Production URL sample: https://ngrok-free.dev/webhook/booking - HTTP Method : POST - Path : booking - Authentication : None - Respond : Using 'Respond to Webhook' Node
9.2 Set - reschedule fields	→ Sets the client name, email, slot for rescheduling and the new booking ID. - Mode : Manual Mapping - Fields to Set - name : clientName, value : {{ \$json.body.clientName }} (String) - name : clientEmail, value : {{ \$json.body.clientEmail }} (String) - name : selectedSlot, value : {{ \$json.body.selectedSlot }} (String) - name : bookingId, value : {{ \$json.body.bookingId }} (String) - Include Other Input Fields : On - Input Fields to Include : All
9.3 Google Calendar - update event	→ Continue to retain the client name and email, plus the new time slot - Credential : Google Calendar account - Resource : Update - Operation : Create - Calendar : From list, primary - Event ID : {{ \$json.bookingId }} - Use Default Reminds : On - Update Fields

	<pre>End : {{ DateTime.fromISO(\$json.selectedSlot).plus({minutes:30}).toISO() }} Start : {{ \$json.selectedSlot }} Summary : Booking (rescheduled) - {{ \$json.clientName }}</pre>
9.4 Set - reschedule bookingid + meetlink	→ Set booking ID, meet link, to send it later • Mode : Manual Mapping • Fields to Set <pre>name : bookingID, value : {{ \$json.id }} (String) name : meetLink, value : {{ \$json.hangoutLink }} (String) name : clientName, value : {{ \$('Set — reschedule fields').item.json.clientName }}(String) name : clientEmail, value : {{ \$('Set — reschedule fields').item.json.clientEmail }}(String) name : selectedSlot, value : {{ \$('Set — reschedule fields').item.json.selectedSlot }}(String)</pre>
9.5 Code - swap book slots	→ Handles when booking ID is changed • Mode : Run Once for All Items • Javascript <pre>const staticData = \$getWorkflowStaticData('global'); const email = \$json.clientEmail; const newSlot = \$json.selectedSlot; const record = staticData[email]; const oldSlot = record ? record.selectedSlot : null; if (!staticData._bookedSlots) staticData._bookedSlots = []; if (oldSlot) { staticData._bookedSlots = staticData._bookedSlots.filter(s => s !== oldSlot); } if (!staticData._bookedSlots.includes(newSlot)) { staticData._bookedSlots.push(newSlot); }</pre>

	<pre> } if (record) { record.selectedSlot = newSlot; record.bookingId = \$json.bookingId; } return [{ json: \$json }]; </pre>
9.6 Respond - reschedule confirmed page	→ This finally says that the booking is successfully rescheduled and automatically edited to calendar. - Respond With : Text - Response Body: Put here the html design for booking update. - Response Headers - Name : Content-Type - Value : text/html; charset=utf-8
9.7 HTML - build updated receipt	→ This is the html design of the pdf that will be used as an updated receipt. Aside from booking details, the header also contains agency name, location, and email. Operation : Generate HTML Template
9.8 Convert to File	→ This defines the content layout and appearance to be passed. - Operation : Convert to Text File - Text Input Field : html - Put Output File in Field : data - Options - File Name : index.html
9.9 HTTP Request Gotenberg	→ This connects to the Gotenberg for pdf generation. - Method : POST - URL : http://<ip address>/forms/chromium/convert/html - Authentication : None

	• Send Body : On • Body Content Type : Form-Data • Body - Type : n8n Binary File - Name : files - Input Data Field Name : data • Response Format : File • Put Output Data : data
9.10 Code - rename PDF	→ This creates a personalized pdf name for the client so it won't generate any pdf name. • Mode : Run Once for All Items • Language : JavaScript • Javascript : <pre>const nameRaw = \$('Set — reschedule bookingId + meetLink').item.json.clientName 'Client'; const parts = nameRaw.trim().split(' '); const firstName = parts[0] 'Firstname'; const lastName = parts.slice(1).join(' ') 'Lastname'; const fileName = `\${lastName}_\${firstName}_Booking.pdf`; const binaryKey = Object.keys(\$binary)[0]; const binaryData = \$binary[binaryKey]; binaryData.fileName = fileName; binaryData.mimeType = 'application/pdf'; return [{ json: \$json, binary: { data: binaryData } }];</pre>
9.11 Gmail - send updated receipt	→ This creates both message and pdf receipt to the email of the client. • Credential: Gmail account 2 • Resource : Message • Operation : Send • To : {{ \$('Set — reschedule bookingId + meetLink').item.json.clientEmail }} • Subject : Your booking has been updated! • Email Type : HTML • Message :

	{{ \$('Set — reschedule bookingId + meetLink').item.json.clientName }}, Your booking has been rescheduled. Your updated receipt is attached. Join Google Meet: {{ \$('Set — reschedule bookingId + meetLink').item.json.meetLink }} Need to cancel or reschedule again? Use the buttons inside the attached PDF. See you soon! - Options : - Attachments : data
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STEPS AFTER SUCCESSFUL BOOKING



Book a Meeting

Enter your details to see available time.

Name

Email

Check Availability

1 of 2

Choose Booking Time

Karl Jung, these are the available slots for the next 7 days.

64 times available

Available times

Confirm Booking

2 of 2



You're all booked!

Your meeting has been confirmed and added to the calendar.

Check your email for your receipt and Google Meet link.

OPENING A BOOKED DOCUMENT

AUTOMATION AGENCY

OFFERING AUTOMATION SOLUTIONS

Makati, Metro Manila, Philippines

+63 (2) 8123-4567

aiagency@gmail.com

BOOKING DETAILS:

Karl Jung

✓ **CONFIRMED** Booking Confirmed

This is the details of your scheduled meeting.

CLIENT NAME	Karl Jung
EMAIL ADDRESS	beauucami@gmail.com
DATE & TIME	Thursday, April 30, 2026 at 09:30 AM
DURATION	30 Minutes
GOOGLE MEET	https://meet.google.com/exb-vhuy-azb
DATE ISSUED	April 29, 2026

NEED TO MAKE CHANGES?

Cancel My Booking

Change My Date

BOOKING ID

k8e0hspmosgd5ccahuae7n0ed4

RESCHEDULING A BOOKING SCHEDULE

To reschedule, pick a new time

Available times

Thu, Apr 30, 09:30 AM

Update Booking

61 times available



Booking Updated!

Your meeting has been moved to the new time and the calendar has been updated.

Check your email for your updated receipt.

FOR BOOKING CANCELLATION



Cancel this booking?

Are you sure you want to cancel? This is a permanent action and can't be undone.

Yes, cancel it

Keep my booking

Your meeting will be permanently removed from the calendar.

RESCHEDULED PDF RECEIPT

AUTOMATION AGENCY
PROFESSIONAL AUTOMATION SOLUTIONS

Makati, Metro Manila, Philippines
+63 (2) 8123-4567
aagency@gmail.com

BOOKING DETAILS: KARL JUNG
Karl Jung

RESCHEDULED

Booking Updated

Your meeting has been moved to a new time.

CLIENT NAME	Karl Jung
EMAIL ADDRESS	beauucami@gmail.com
NEW DATE & TIME	Thursday, April 30, 2026 at 10:00 AM
DURATION	30 Minutes
GOOGLE MEET	https://meet.google.com/exb-vhuy-azb
DATE UPDATED	April 29, 2026

NEED TO MAKE CHANGES AGAIN?

Cancel My Booking

Change My Date

BOOKING ID
k8e0hspmosgd5ccahuae7n0ed4

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 Booking is Not Cancelled Make sure to be there. We look forward to seeing you!	 Booking Cancelled Your booking has been cancelled.